

BRUNO PAOLO BARBARAN QUIROZ

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PROFILE

Bachelor in Software Engineering from Universidad Tecnológica del Perú, with experience in frontend and backend development, specialized in technologies such as React. I have an intermediate level of English and a great passion for technology, with a focus on artificial intelligence and web development. I have worked on natural language processing and application optimization projects. In addition, I have a certificate as a STEM tutor in Algorithmics and I was an active member of the Microsoft UTP community.

EXPERIENCE

CECOM San Juan de Lurigancho

Sep 2024 – Present

Communications Center focused on the modernization of public management and the improvement of municipal services through technology.

Development Area Coordinator

- Implemented a comprehensive system for attendance control and personnel management, human resources administration.
- I developed and implemented a tracking map for Tetra radios, incorporating a real-time alerting algorithm that notifies if they leave assigned areas or exceed speed, improving operational monitoring and safety.
- I designed and deployed a citizen web platform to validate the identity and authorization of municipal personnel, promoting transparency and trust in public services.
- I supervised the maintenance and continuous improvement of municipal applications, ensuring their operability, technological updating and adaptability to new needs.

Dominiotech

Oct 2022 – Dic 2023

Information Technology Company for the optimization of environmental and occupational safety commitments.

Web Developer

- I implemented a system with Google Maps that reduced the time to locate the monitoring stations by 30%.
- I created a virtual assistant with artificial intelligence through natural language processing that improved internal document search efficiency by 40%, streamlining administrative processes.
- I designed a map for meteorological monitoring that allowed a 15% reduction in operating costs thanks to a better planning of activities.
- I developed a platform for visualizing leave results that increased team productivity by 25% by facilitating data access and analysis.
- I designed a Stiff diagram map for water monitoring, improving the visualization and analysis of water quality.

PUBLICATIONS

"Implementation of a Natural Language Processing Model for Spanish-Quechua Machine Translation".
Lecture Notes in Networks and Systems (LNNS), 8th Computational Methods in Systems and Software 2024.

- **Authors:** B. Barbaran y W. Ticona.
- **Description:** Research focused on developing and optimizing translations between Spanish and Quechua, using the natural language processing (NLP) model Helsinki-XNLP/opus-mt- es-en.
- **Methodology:** Application of advanced hyperparameter optimization techniques using Optuna, achieving significant improvements in the accuracy of the translation model.
- **Publisher:** Springer Nature

- **Status:** Article in process of publication in a Scopus indexed journal.

SKILLS

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|---------|-----------|--------------|
| • HTML | • CSS | • JavaScript |
| • React | • Node JS | • Python |
| • SQL | • Java | • Flask |

EDUCATION

Universidad Tecnológica del Perú
Software Engineering.

Mar 2019 - 2023

ADDITIONAL INFORMATION

- STEM Tutor - Algorithmic.
- Member of the Microsoft UTP community.
- Intermediate Excel certificate.